



NYU

**TANDON SCHOOL
OF ENGINEERING**

Integrated Design & Media

Course Number and Section, Course Title

INSTRUCTOR INFORMATION

- Thomas Martinez
- 370 Jay Street
- 310.951.3985
- Monday 10AM-12PM
- thomas.martinez@nyu.edu

COURSE INFORMATION

- DM-UY 2193 B
- Intro to Web
- This class introduces the web as a creative medium. Using HTML, CSS and JavaScript participants will create personal archives, poetic reflections, and works of interactive multimedia for desktop and mobile viewing. Through our practice, students will come away with technical strategies for making modern, responsive, and accessible websites. We will also gain various historical perspectives on the web and the many ways it has been used by artists, designers, and activists. Coursework will take the form of creative, and technical exercises, with some assigned readings and reflections. Guiding our practice will be an attunement to our own creative voice and the ways it can take shape in the web browser. The class will end with a showcase of student websites.
- 3 Credits, Twice a Week
- Tuesday and Thursday 10AM-11:50AM
- 370 Jay St Room 310 Loc: Brooklyn Campus

COURSE OVERVIEW AND GOALS

In this class we'll cover the basics of what are known as a "front-end" web stack including HTML, CSS, and JavaScript. We'll also touch on some back-end or "server-side" programming

and databases as well. Students will be able to build efficient, responsive websites that draw on personal experiences or from our own creative practice. The goal of the class is to help you achieve fluency and expressivity in web-based programming environments while developing your own unique style. You will also understand how the internet works, its infrastructure, and gain a historical and social perspective through lectures and readings.

UPON COMPLETION OF THIS COURSE, STUDENTS WILL BE ABLE TO:

- Make a website using HTML, CSS, and JavaScript
- Use version control and collaborative coding tools like GitHub
- Access databases using Web APIs
- Create responsive and accessible user interfaces
- Use a variety of multimedia libraries built for the web

COURSE REQUIREMENTS

CLASS PARTICIPATION

I am committed to facilitating the fullest possible participation of all students. There are many forms of participation. Please communicate what kinds of engagement are best for you so it can be taken into account.

Examples of modes of participation can look like: asking questions, going to office hours, sending and reading emails, class group discussion, arriving on time, going to class, taking notes, listening to peers, submitting responses to a form (anonymous or not), following instructions, active listening, etc.

GRADING OF ASSIGNMENTS

The grade for this course will be determined according to the following:

Assignments/Activities	% of Final Grade
Class participation	20%
Attendance	20%
Weekly Assignments	40%
Final Project	20%

LETTER GRADES

Letter grades for the entire course will be assigned as follows:

Letter Grade	Points	Percent
A	4.00	Example: 92.5% and higher
A-	3.67	Example: 90.0 – 92.49%
B+	3.33	Example: 87.5% - 89.99%

B	3.00	Example: 82.5% - 87.49%
B-	2.67	Example: 80% - 82.49%
C+	2.33	Example: 77.5% - 79.99%
C	2.00	Example: 72.5% - 77.49%
C-	1.67	Example: 70% - 72.49%
D+	1.33	Example: 67.5% - 69.99%
D	1.00	Example: 62.5% - 67.49%
D-	.67	Example: 60% - 62.49%
F	.00	Example: 59.99% and lower

HOW TO ACCESS YOUR GRADES

Grades are accessible via Albert.

COURSE SCHEDULE

TOPICS AND ASSIGNMENTS

Week/Date	Topic	Reading	Assignment Due
Week 1: Intro to web , January 20th, 22nd	Syllabus Basic HTML, text-editors, Hypertext Narratives	Chapter 1 Learning Web Design	Zine or Hypertext Narrative

Week 2: CSS , January 27, 29th	Stylesheets, Fonts, Colors, General Design Guidelines	Chapter 2A Learning Web Design	CSS still life
Week 3: Deployment, February 3rd, 5th	GitHub, hosting, domain names	n/a	Deploy to the Web
Week 4: Divs and Responsiveness, February 10th, 12th	Flexboxes, Media Screen, More CSS 😊	Chapter 2B Learning Web Design	Make your Zine Responsive
Week 5: Scripting the Web, February 17th, 19th	Basic Programming Techniques, DOM Interaction, Scripting Media Attributes, Algorithms	Chapter 3 Learning Web Design	An procedurally generated website, ambient website
Week 6: Web APIs, February 24th 26th	Querying External Data	n/a	Build a site that uses weather data (or another API)
Week 7: Device APIs, March 3rd, 5th	Geolocation and "site specific" websites.	n/a	Build a site specific website, or an "instrument"
Week 8: Web Audio, March 10th, 12th	Vanilla Web Audio API, Synthesizers and Audio Playback	n/a	Make a song
Week 9: Canvas APIs, March 17th, 19th	Create custom GUIs and Animations using the Vanilla Canvas API	n/a	Make a painting
Week 10: Libraries, March 24th, March 26th	p5.js, p5.sound.js ml5.js,	n/a	Make an interactive sound work or "gestural instrument"
Week 11: Libraries, March 31st, April 2nd	Howler.js, Three.js, A-Frame	n/a	Final Project Proposal

Week 12: Databases and Node, April 7th, 9th	Spreadsheets as a Database, Making Networked Applications	n/a	Extra Credit: Build a website you can update with a google sheet, or make a networked application.
Week 13: Making a Library, April 14th, 16th	Rollup, Build Process, Hosting your own libraries	n/a	Extra Credit: Build your own Library :) from scratch. eg random fact generator, translation, or accessibility library
Week 14: Final Project Workshop, April 21st, 23rd	Work on technical challenges for final projects	n/a	Bring in projects to work on
Week 15: Final Project Workshops and Presentations, April 28th, 30th	In class presentations, Code Reviews	n/a	Group 1 Present Projects
Week 16: Final Project Presentations, May 5th	In class presentations, Code Reviews	n/a	Group 1 Present Projects

EXPECTATIONS FOR WORK OUTSIDE THE CLASSROOM

Students should expect to spend roughly 5 hours each week on supplemental work in this course. This may include reading assignments, writing, exam preparation, research, homework assignments, building, writing code, study time, unsupervised lab work, unsupervised group work, etc.

REQUIRED TEXTBOOKS & MATERIALS

- Learning Web Design, 6th Ed. by Jeniffer Robbins

RESOURCES

- **Access your course materials:** [LINK TO YOUR COURSE WEBSITE]
- **Databases, journal articles, and more:** [Dibner Library](https://library.nyu.edu) (library.nyu.edu)
- **Assistance with strengthening your writing:** [NYU Writing Center](https://nyu.mywconline.com) (nyu.mywconline.com)
- **Obtain 24/7 technology assistance:** [IT Service Desk \(NYU IT\)](https://nyu.edu/it/servicedesk) (nyu.edu/it/servicedesk)

COURSE POLICIES

ATTENDANCE AND TARDINESS

You may have 4 excused absences before your grade is affected. Communication is preferred in the case of absences so that I can work with you on missed assignments and catching up.

LATE ASSIGNMENT

Late assignments may be turned in past the due date for a maximum of 80% credit.

ACADEMIC HONESTY/PLAGIARISM

[Insert Your Policy Here and keep the following]

NYU School of Engineering Policies and Procedures on Academic Misconduct (from the School of Engineering Student Code of Conduct)

Introduction: The School of Engineering encourages academic excellence in an environment that promotes honesty, integrity, and fairness, and students at the School of Engineering are expected to exhibit those qualities in their academic work. It is through the process of submitting their own work and receiving honest feedback on that work that students may progress academically. Any act of academic dishonesty is seen as an attack upon the School and will not be tolerated. Furthermore, those who breach the School's rules on academic integrity will be sanctioned under this Policy. Students are responsible for familiarizing themselves with the School's Policy on Academic Misconduct.

Definition: Academic dishonesty may include misrepresentation, deception, dishonesty, or any act of falsification committed by a student to influence a grade or other academic evaluation. Academic dishonesty also includes intentionally damaging the academic work of others or assisting other students in acts of dishonesty. Common examples of academically dishonest behavior include, but are not limited to, the following:

- *Cheating:* intentionally using or attempting to use unauthorized notes, books, electronic media, or electronic communications in an exam; talking with fellow students or looking at another person's work during an exam; submitting work prepared in advance for an in-class examination; having someone take an exam for you or taking an exam for someone else; violating other rules governing the administration of examinations.
- *Fabrication:* including but not limited to, falsifying experimental data and/or citations.
- *Plagiarism:* intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise; failure to attribute direct quotations, paraphrases, or borrowed facts or information.
- *Unauthorized collaboration:* working together on work that was meant to be done individually.

- *Duplicating work*: presenting for grading the same work for more than one project or in more than one class, unless express and prior permission has been received from the course instructor(s) or research adviser involved.
- *Forgery*: altering any academic document, including, but not limited to, academic records, admissions materials, or medical excuses.

Access the entire School of Engineering Student Code of Conduct here:
engineering.nyu.edu/academics/code-of-conduct

GENERATIVE TOOL USE IN THIS CLASS

Far be it from me to say “You Cannot Use AI for your Assignments.” You’ll use it, and it can be an excellent aid; but because this is a class where you are expected to gain some self-sufficiency in web-programming, I’ll always favor simple, eloquent code over tons and tons of complex AI generated stuff. Oftentimes, that code is hard to parse, it will be difficult for me to troubleshoot it for you, and in general, if you are unable to explain it, you won’t be learning much.

I use it, for full transparency, and you’ll even see me using it during class at times, but there should be a level of understanding when using it.

Document Your Process And Use Of Generative AI When Completing Assignments Using Generative AI

- Save copies of each step to create a record that can be shared with your instructors to facilitate conversations about your work. For example, keep copies of your previous drafts before and after interacting with generative AI.

Ensure that your final work is your own and is not simply copied and pasted from a generative AI tool

- Your own style and voice should be evident.
- Simply rephrasing AI-generated content is not enough for it to be considered your own work! You must still apply your own critical and creative thinking to ensure learning.
- Foreground [intellectual virtues and beware of cognitive biases](#) from the [Open Inquiry Toolkit](#).

Fact-check and Cross-Verify Any Information You Use From Generative AI

- Note the [limitations of generative AI](#), most notably that content generated by AI may be biased, made up, inaccurate, not up to date, etc.
- Apply critical thinking at all times! It is important to fact-check and cross-verify any information generative AI gives you.

- Using generative AI is not the same as using a search engine.

Think for yourself

- Form your own perspectives and points of view. Do not rely solely on information generated by generative AI tools.

ACADEMIC ACCOMMODATIONS

If you are a student with a disability who is requesting accommodations, please contact New York University's Moses Center for Student Accessibility (CSA) at 212-998-4980 or mosescsa@nyu.edu. You must be registered with CSA to receive accommodations. Information about the Moses Center can be found at <https://www.nyu.edu/csa>. The Moses Center is located at 726 Broadway on the 2nd floor.

If you are experiencing an illness or any other situation that might affect your academic performance in a class, please email the Office of Advocacy, Compliance and Student Affairs: eng.studentadvocate@nyu.edu.

STATEMENT ON INCLUSION

The NYU Tandon School values an inclusive and equitable environment for all our students. I hope to foster a sense of community in this class and consider it a place where individuals of all backgrounds, beliefs, ethnicities, national origins, gender identities, sexual orientations, religious and political affiliations, and abilities will be treated with respect. It is my intent that all students' learning needs be addressed, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. If this standard is not being upheld, please feel free to speak with me.

RESOURCES FOR NON-CITIZEN STUDENTS

More than 40 percent of NYU students are international students. A smaller number are undocumented students, but many more come from mixed status families and communities. As a professor, I am committed to doing everything I can to ensure that every student, regardless of immigration status, is safe in this classroom. Following the recommendation of [the NYU chapter of the AAUP](#), I encourage students to seek free legal support and other resources through NYU's [Immigrant Defense Initiative](#). NYU IDI provides an [extensive list of updates and resources](#). Students may also consult the "[Know Your Rights](#)" information provided by the New York Immigration Coalition.